

# **Prompt Engineering Techniques with Examples**

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## **• What is Prompt Engineering?**

Prompt Engineering is the process of designing and structuring prompts (instructions or questions) given to Artificial Intelligence systems in order to generate accurate, relevant, and meaningful responses. It helps users communicate effectively with AI tools such as ChatGPT, Gemini, Claude, and Midjourney.

A prompt can be a question, command, statement, or even a partially completed sentence. The quality of the prompt directly affects the quality of the AI response.

## **• Importance of Prompt Engineering**

- ☐ Improves the accuracy of AI responses.
- ☐ Helps generate detailed and context-aware answers.
- ☐ Saves time by reducing repeated instructions.
- ☐ Enables AI to perform specialized tasks.

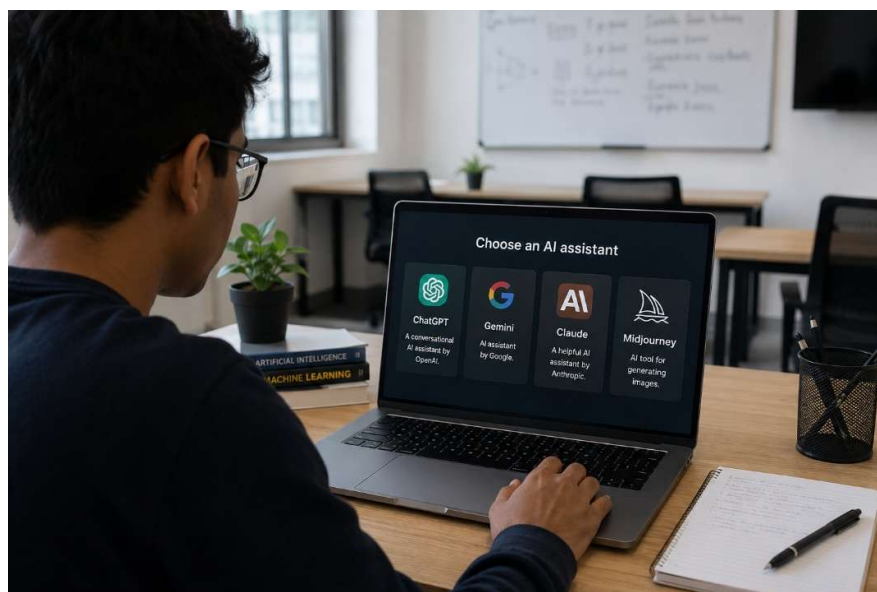
- ☐ Useful in education, business, coding, research, and content creation.

## Common AI Tools Used

- ☐ ChatGPT
- ☐ Gemini
- ☐ Claude
- ☐ Midjourney
- ☐ Microsoft Copilot

## Applications of Prompt Engineering

- ☐ Text Summarization
- ☐ Content Writing
- ☐ Programming Assistance
- ☐ Image Generation
- ☐ Data Analysis
- ☐ Educational Support
- ☐ Research and Documentation



## **Components of a Good Prompt**

### **1. Instruction**

Clearly tells the AI what task to perform. Example: “Explain the causes of World War II.”

### **2. Context**

Provides background information. Example: “Explain the causes of World War II for school students.”

### **3. Input Data**

Gives specific content or information. Example: “Summarize the following paragraph in 5 lines.”

### **4. Output Format**

Specifies how the response should appear. Example: “Provide the answer in tabular format.”

### **5. Constraints**

Adds limitations or conditions. Example: “Explain in less than 100 words.”

## **Characteristics of Good Prompts**

- Clear and specific
- Simple language
- Context-based
- Goal-oriented
- Structured properly

Example

Prompt:

“Explain the Solar System in simple words for class 6 students in bullet points.”

Response:

The AI gives a simplified explanation using easy words and bullet points.

## **Straightforward Prompting**

Definition

Straightforward Prompting is the simplest type of prompting where the user directly asks a question or gives a command to the AI.

The AI responds immediately based on the given request.

Features

- Simple and direct
- Easy to use
- Best for general questions
- Requires minimal context

Example

**Prompt:**

“When did World War II start?”

**Response:**

World War II started on September 1, 1939, when Germany invaded Poland.

**Advantages**

- Fast response generation
- Easy for beginners
- Useful for factual information

## **Disadvantages**

- May produce vague answers
- Limited context awareness
- Less effective for complex tasks

## **Uses**

- Quick questions
- General knowledge
- Simple calculations
- Basic explanations

## **Tabular Format Prompting**

### Definition

Tabular Format Prompting is a technique where the user asks the AI to provide information in the form of a table.

This method improves readability and comparison.

### Features

- Organizes information clearly
- Easy to compare data
- Improves presentation quality
- Useful for educational purposes

### Example

Prompt:

“Tabulate the causes and effects of World War II.”

Causes

Effects

Treaty of Versailles

Massive destruction

Rise of dictators

Millions of deaths

Economic depression

Formation of United Nations

## **Preceding Question Prompting**

### Definition

Preceding Question Prompting involves asking a main question followed by additional related questions to guide the AI toward a more detailed response.

This technique improves contextual understanding.

### Features

- Produces detailed responses
- Improves logical flow
- Encourages structured explanations
- Useful for long answers

### Example 1

#### Prompt:

“Explain the Quit India Movement. Explain its effects in India.”

#### Response Includes:

- Introduction
- Background
- Launch of the movement
- Effects on India

### **Advantages**

- Generates detailed explanations
- Better understanding of topics
- Structured output

### **Applications**

- Essay writing
- Research work
- Educational assignments
- Report preparation

## **Missing Word Prompting**

### Definition

Missing Word Prompting is a technique where one or more words are intentionally left blank for the AI to complete.

This technique is commonly used in educational exercises and language learning.

### Features

- Interactive learning method
- Helps improve memory
- Useful for quizzes and exams
- Encourages prediction capability

## **Advanced Prompt Engineering Techniques**

### 1. Role-Based Prompting

The user assigns a role to the AI.

Example:

“You are a financial advisor. Explain compound interest.”

Benefit:

AI provides specialized and professional responses.

## 2. Step-by-Step Prompting

The AI is asked to solve problems in sequence.

Example:

“Solve the math problem step-by-step.”

Benefit:

Improves logical reasoning and understanding.

## 3. Chain-of-Thought Prompting

Encourages the AI to explain reasoning before giving the answer.

Example:

“Think step-by-step and calculate the speed.”

## 4. Constraint-Based Prompting

Adds restrictions to the response.

Example:

“Explain Artificial Intelligence in 50 words.”

# **Benefits and Challenges of Prompt Engineering**

Benefits

## 1. Better AI Communication

Users can interact more effectively with AI systems.

## 2. Improved Productivity

Saves time in writing, coding, and research.



### 3. Personalized Responses

AI can generate customized outputs.

### 4. Enhanced Creativity

Useful for storytelling, design, and brainstorming.

### 5. Educational Support

Helps students understand complex concepts.

### Challenges

#### 1. Ambiguous Prompts

Unclear prompts may generate incorrect responses.

#### 2. Bias in AI Output

AI responses may contain bias depending on training data.

#### 3. Overdependence on AI

Excessive use may reduce critical thinking skills.

#### 4. Need for Practice

Effective prompting requires experimentation and learning.

| <div><h2>Prompt Engineering Techniques</h2><p>with Examples</p><p>Prompt engineering is the art and science of designing effective prompts to get accurate, relevant and high-quality responses from AI models.</p></div>                                                                                                                                                                      |                                                                           |                                                                                                                                            |                                                                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <div> ChatGPT Gemini Claude Midjourney</div> |                                                                           |                                                                                                                                            |                                                                                                                      |
| Technique                                                                                                                                                                                                                                                                                                                                                                                      | Description                                                               | Example Prompt                                                                                                                             | Example Output (Summary)                                                                                             |
|  <b>1. Zero-Shot Prompting</b>                                                                                                                                                                                                                                                                              | Ask the model to perform a task without providing examples.               | "Explain the benefits of renewable energy."                                                                                                | Renewable energy reduces carbon emissions, is sustainable, lowers long-term costs, and improves energy security.     |
|  <b>2. Few-Shot Prompting</b>                                                                                                                                                                                                                                                                               | Provide a few examples to show the model the desired pattern.             | "Q: What is the capital of France?<br>A: Paris<br>Q: What is the capital of Japan?<br>A: Tokyo<br>Q: What is the capital of India?<br>A: " | The capital of India is New Delhi.                                                                                   |
|  <b>3. Role-Based Prompting</b>                                                                                                                                                                                                                                                                             | Assign a role to the model to get a more focused and contextual response. | "You are a financial advisor. Explain how compound interest works in simple terms."                                                        | Compound interest is earning interest on both your initial investment and the interest that accumulates over time... |
|  <b>4. Step-by-Step Prompting</b>                                                                                                                                                                                                                                                                           | Ask the model to solve a problem step-by-step.                            | "Solve the following math problem step-by-step:<br>$23 + 47 \times 2$ "                                                                    | 1. Multiply 47 by 2 = 94<br>2. Add 23 to 94 = 117<br>Answer: 117                                                     |
|  <b>5. Output Format Prompting</b>                                                                                                                                                                                                                                                                          | Specify the desired format for the output.                                | "List 3 popular programming languages in JSON format."                                                                                     | <pre>{<br/>  "languages": ["Python", "JavaScript", "Java"]<br/>}</pre>                                               |
|  <b>6. Chain-of-Thought Prompting</b>                                                                                                                                                                                                                                                                       | Encourage the model to think step-by-step before giving the final answer. | "If a train takes 3 hours to travel 240 km, how long will it take to travel 400 km?<br>Let's think step-by-step."                          | 1. Speed = $240 / 3 = 80$ km/h<br>2. Time = $400 / 80 = 5$ hours<br>Answer: 5 hours                                  |
|  <b>7. Constraint-Based Prompting</b>                                                                                                                                                                                                                                                                       | Add constraints to guide the model's response.                            | "Write a short story about AI in 100 words or less."                                                                                       | (A 100-word short story about AI...)                                                                                 |
| <div> <b>Best Practices:</b> • Be clear and specific • Provide context • Use examples • Break complex tasks into steps • Iterate and refine prompts</div>                                                                                                                                                   |                                                                           |                                                                                                                                            |                                                                                                                      |

## **Conclusion**

Prompt Engineering plays an important role in improving communication between humans and Artificial Intelligence systems. By using proper prompting techniques such as Straightforward Prompting, Tabular Prompting, Preceding Question Prompting, and Missing Word Prompting, users can obtain more accurate and meaningful responses.

Advanced techniques like Role-Based Prompting and Chain-of-Thought Prompting further improve the quality of AI-generated outputs. Prompt Engineering is becoming an essential skill in education, business, research, software development, and content creation.

As AI technologies continue to grow, learning effective prompt engineering techniques will help users maximize the power of AI tools such as ChatGPT, Gemini, Claude, and Midjourney.